BRITISH STANDARD

BS EN 14790:2005

Stationary source emissions — Determination of the water vapour in ducts

The European Standard EN 14790:2005 has the status of a British Standard

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British Standards

National foreword

This British Standard is the official English language version of EN 14790:2005.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/1, Stationary source emissions, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep UK interests informed;
- monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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EUROPEAN STANDARD

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EN 14790

November 2005

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English Version

Stationary source emissions - Determination of the water vapour in ducts

Emissions de sources fixes - Détermination de la vapeur d'eau dans les conduits

Emissionen aus stationären Quellen - Bestimmung von Wasserdampf in Leitungen

This European Standard was approved by CEN on 30 September 2005.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

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Foreword

This European Standard (EN 14790:2005) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2006, and conflicting national standards shall be withdrawn at the latest by May 2006.

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this European Standard.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This European Standard describes the condensation/adsorption technique, including the sampling system, to determine the water vapour concentration in the flue gases emitting to atmosphere from ducts and stacks.

This technique is usually used all over Europe for water vapour monitoring. However to be implemented as the Standard Reference Method (SRM), the user has to demonstrate that the performance characteristics of the method are better than the performance criteria defined in this European Standard and that the overall uncertainty of the method is less than ± 20 % of the measured value. This European Standard as the Standard Reference Method (SRM) is used for periodic monitoring and for calibration or control of Automatic Measuring Systems (AMS) permanently installed on a stack, for regulatory purposes or other purposes.

An Alternative Method to this SRM may be used provided that the user can demonstrate equivalence according to the Technical Specification CEN/TS 14793, to the satisfaction of his national accreditation body or law.

The determination of water vapour is mainly necessary for:

— regulatory purposes, to express the concentration at standard conditions (on dry gas);

— adjust the flow rate for isokinetic sampling, when a dry gas flow rate metering device is used.

For both applications, the quantity to be measured is the amount of water present in the gas phase (vapour), which does not include water droplets.

This European Standard is applicable in the range from 4 % to 40 % relative humidity and for water vapour concentration from 29 g/m^3 to 250 g/m^3 as a wet gas, although for a given temperature the upper limit of the method is related to the maximum pressure of water in air or in the gas.

This European Standard has been evaluated during field tests on waste incineration, co-incineration and large combustion installations. It has been validated for sampling periods of 30 min in the concentration range of 7 % to 26 % volume.

In this European Standard all the concentrations are expressed in normal conditions (273 K and 101,3 kPa).

NOTE For saturated conditions the condensation/adsorption method is not applicable. Some guidance is given in this European Standard to deal with flue gas when droplets are present.

2 Normative references

The following referenced documents are indispensable for the application of this European Standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainly in measurement*.

CEN/TS 14793, *Stationary source emission - Intralaboratory validation procedure for an alternative method compared to a reference method*.

EN ISO 14956, *Air Quality – Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*.

3 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

3.1
absorber
device in which water vapour is absorbed

3.2
detection limit (L_D)
concentration value of the measurand below which there is at least 95 % level of confidence that the measured value corresponds to a sample free of that measurand

3.3
dew point
temperature below which the condensation of water vapour begins at the given pressure condition of the flue gas

3.4
droplets
small liquid particles of condensed water vapour or water liquid in the flue gas (e.g. coming from a scrubber)

NOTE In adiabatic equilibrium conditions, droplets could arise only if a gas stream is saturated with water.

3.5
measurand
particular quantity subject to measurement

[VIM 2.6]

3.6
measuring series
several successive measurements carried out at the same sampling plane and with the same process operating conditions

[EN 13284-1]

3.7
repeatability in the laboratory
closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement

NOTE 1 Repeatability conditions include:

— same measurement procedure;

— same laboratory;

— same sampling equipment, used under the same conditions;

— same location;

— repetition over a short period of time.

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability is expressed as a value with a level of confidence of 95 %.

[VIM 3.6]

3.8
repeatability in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with two equipments under the same conditions of measurement

NOTE 1 These conditions include:

— same measurement procedure;

— two equipments, the performances of which are fulfilling the requirements of the reference method, used under the same conditions;

— same location;

— implemented by the same laboratory;

— typically calculated on short periods of time in order to avoid the effect of changes of influence parameters (e.g. 30 min).

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability under field conditions is expressed as a value with a level of confidence of 95 %.

3.9
reproducibility in the field
closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with several equipments under the same conditions of measurement

NOTE 1 These conditions include:

— same measurement procedure;

— several equipments, the performances of which are fulfilling the requirements of the reference method, used under the same conditions;

— same location;

— implemented by several laboratories.

NOTE 2 Reproducibility may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.10
sampling location
specific area close to the sampling plane where the measurement devices are set up

3.11
sampling plane
plane normal to the centreline of the duct at the sampling position

[EN 13284-1]

3.12
sampling point
specific position on a sampling line at which a sample is extracted

[EN 13284-1]

3.13
standard reference method (SRM)
measurement method recognised by experts and taken as a reference by convention, which gives, or is presumed to give, the accepted reference value of the concentration of the measurand (3.5) to be measured

3.14
uncertainty
parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

3.14.1
standard uncertainty u
uncertainty of the result of a measurement expressed as a standard deviation u

3.14.2
expanded uncertainty U
quantity defining a level of confidence about the result of a measurement that may be expected to encompass a specific fraction of the distribution of values that could reasonably be attributed to a measurand

$U = k \times u$

NOTE In this European Standard, the expanded uncertainty is calculated with a coverage factor of k = 2, and with a level of confidence of 95 %.

3.14.3
combined uncertainty u_c
standard uncertainty uc attached to the measurement result calculated by combination of several standard uncertainties according to GUM

3.14.4
overall uncertainty U_c
expanded combined standard uncertainty attached to the measurement result calculated according to GUM

$U_c = k \times u_c$

3.15
uncertainty budget
calculation table combining all the sources of uncertainty according to ISO 14956 or ENV 13005 in order to calculate the overall uncertainty of the method at a specified value

3.16
vapour pressure
pressure of water in vapour form

4 Principle

4.1 General

This European Standard describes the Standard Reference Method (SRM) for determining water vapour content emitting to atmosphere from ducts and stacks. The specific components and the requirements for the measuring system are described. A number of performance characteristics, together with associated minimum performance criteria are specified for the measurement method (see Table 1 in 7.2). The overall uncertainty of the method shall meet the specifications given in this European Standard.

The method described hereafter is appropriate when the flue gas is free of droplets.

Within the scope of this European Standard, it is assumed that gas streams in stacks or ducts are more or less in adiabatic (thermodynamic) equilibrium. In those conditions, droplets can arise only if a gas stream is saturated with water. When no droplets are present in the gas stream, the gas stream is then assumed to be unsaturated with water. A gas sample is extracted at a constant rate from the stack. The water vapour of that sample is subsequently trapped by adsorption or by condensation plus adsorption; the mass of the vapour is then determined by weighing the mass gain of the trapping unit.

When droplets are present in the gas stream, the implementation of the method described in this European Standard leads to an overestimation of the water vapour content. If the measured value is equal to or higher than the expected value shown in the table in Annex A for saturated conditions at the temperature and pressure of the flue gas, that means that the presence of droplets can lead to biased results; such results shall be rejected.

In such cases, the evidence suggests that the gas stream is saturated with water vapour. Under these conditions, the method is abridged to a determination of the gas temperature. Then, the water vapour concentration is calculated from the theoretical mass of water vapour per unit of standard gas volume at liquid-gas equilibrium, given the actual temperature, pressure and composition of the gas stream.

4.2 Adsorption or condensation/adsorption method

A measured quantity of sampled gas is extracted from the gas stream through a trapping unit, which meets the specifications of efficiency (see 6.4.2). The mass gain of the trapping unit is measured and divided by the volume sampled in order to determine the mass concentration of water vapour.

4.3 Temperature method

This method applies when gases are water saturated.

A temperature probe is placed in the gas stream saturated with water vapour, until it reaches equilibrium. The amount of water vapour present in the gas is subsequently derived from the temperature, using a water liquid-gas equilibrium chart or table (see Annex A).

5 Apparatus

5.1 General

A known volume of flue gas is extracted representatively from a duct or chimney during a certain period of time at a controlled flow rate. A filter removes the dust in the sampled volume; thereafter the gas stream is passed through a trapping unit. It is important that all parts of the sampling equipment upstream of the trapping unit are heated and that the components shall not react with or absorb water vapour (e.g. stainless steel, borosilicate glass, quartz glass, PTFE or titanium are suitable materials).

An example of suitable sampling trains are shown in Annex B. The user can choose between a trapping unit made up with either:

— adsorption unit (Figure B.1); or

— condensation and an adsorption unit (Figure B.2).

The choice shall be made to fulfil the efficiency that is required in 6.4.2.

5.2 Sampling probe

In order to access the representative measurement point(s) of the sampling plane, probes of different lengths and inner diameters may be used. The design and configuration of the probe used shall ensure the residence time of the sample gas within the probe is minimised.

The procedure of clause 6.2.2 can be used when the operator suspects that the flue gas is inhomogeneous.

The probe may be marked before sampling in order to demonstrate that the representative measurement point(s) in the measurement plane has (have) been reached.

The sampling probe shall be surrounded by a heating jacket capable of producing a controlled temperature of at least 120 °C and 20 °C higher than the (acid) dew point of gases and shall be protected and positioned using an outer tube.

NOTE 1 It is possible to perform the sampling of SO_2 and water vapour simultaneously with the same probe (without nozzle providing no droplets are present).

NOTE 2 It is possible to perform the sampling of HCl and water vapour simultaneously with the same probe (without nozzle providing no droplets are present).

5.3 Filter housing

The filter housing shall be made of materials inert to water vapour and shall have the possibility to be connected with the probe thereby avoiding leaks.

The filter housing may be located either:

— in the duct or chimney, mounted directly behind the entry nozzle (in-stack filtration); or

— outside the duct or chimney, mounted directly behind the suction tube (out-stack filtration).

The filter holder shall be connected to the probe without any cold path between the two.

NOTE In special cases where the sample gas temperature is > 200 °C, the heating jacket around the sampling probe, filter holder and connector fine may be omitted. However the temperature in the sampled gas just after the filter housing should not fall below the acid dew point temperature.

5.4 Particle filter

Particle filters and filter housings of different designs may be used, but the residence time of the sample gas should be minimised.

5.5 Trapping unit

The trapping unit shall be made up with:

— adsorption unit; or

— condensation and an adsorption unit.

a) When using the adsorption unit alone, it shall consist of at least one cartridge, impinger or absorber, filled with a suitable drying agent, for example: coloured silica gel.

b) Condensation and adsorption unit shall consist of two stages:

  1) the first one shall be a condensation stage with an optional cooling system;

  2) the second one shall be an adsorption stage as described in a).

The temperature at the outlet of the condensation unit shall be as low as possible.

The efficiency of the sampling system shall be checked according to the procedure described in 6.4.2.

NOTE The trapping efficiency can be increased by increasing the residence time of sampled gases in the trapping unit and/or by improving the efficiency of the cooling system. The sampled volume should be sufficient to reach an appropriate accuracy of the measurement (see clause 5.8 and 7).

Condensation of water shall be avoided in all parts of the sampling system that are not weighed.

5.6 Cooling System (optional)

Any kind of cooling system may be used to condense water vapour in the sampled flue gas (e.g. crushed ice or cryogenic system).

5.7 Sampling pump

A leak-free pump capable of drawing sample gas at a set flow-rate is required.

NOTE 1 A rotameter (optional) could make easier the adjustment of the nominal sampling flow-rate.

NOTE 2 A small surge tank may be used between the pump and rotameter to eliminate the pulsation effect of the diaphragm pump on the rotameter.

NOTE 3 A regulating valve (optional) would also be useful for adjusting the sample gas flow-rate.

5.8 Gas volume meter

Two variants of gas volume meter may be used:

— dry-gas volume meter; or

— wet-gas volume meter.

a) Gas volume meter (wet or dry) shall have a relative uncertainty not exceeding ± 2,0 % of the measured volume (actual conditions).

The gas volume meter shall be equipped with a temperature measuring device (uncertainty less than ± 2,5 K) and shall be associated to an absolute pressure measurement (uncertainty less than ± 1,0 %).

b) When using a dry gas volume meter, a condenser and/or a gas drying system shall be used which can achieve a residual water vapour content of less than 10,0 g/m^3 (equivalent to a dew point of 10,5 °C or a volume content $\chi(H_2O)$ = 1,25 %).

NOTE For example, a glass cartridge or adsorption bottle packed with silica gel (1 mm to 3 mm particle size) which has been previously dried at least at 110 °C for at least 2 h.

When using a wet gas volume meter, a correction shall be applied for water vapour, using table of Annex A.

5.9 Barometer

Barometer capable of measuring atmospheric pressure present at the sampling location, with an uncertainty that does not exceed ± 1 kPa.

5.10 Balance

The resolution of the balance shall be equal or better than 0,1 g or 2,0 % of the water weight to be measured.

5.11 Temperature measurement

Calibrated thermometer for flue gas temperature determination with a measuring range 273 K – 373 K, with an uncertainty of ± 2,5 K or better. That thermometer shall have a low thermal inertia, in order to be rapidly in thermal equilibrium with the stack gas.

6 Measurement procedure

6.1 General requirements

The sample and the point of measurement shall be representative of the emission of the process.

The following points should be considered when sampling:

— nature of the plant process e.g. steady state or discontinuous. If possible the sampling and measuring programme should be carried out under steady operating conditions at the plant;

— expected concentration to be measured and any required averaging period, both of which can influence the measuring and sampling time.

The minimum sampling period is 30 min and the minimum sample gas volume is 50 l. When the expected mass concentration of water vapour is low, the sampling period or the sampling volume may be extended.

6.2 Preparation and installation of equipment

6.2.1 Sampling location

The sampling location chosen for the measurement devices and samplings shall be of sufficient size, easily passable and made in such a way that an emission measurement representative for the measurement task and technically perfect is possible. In addition, the sampling location shall be chosen with regard to safety of the personnel, accessibility and availability of electrical power.

6.2.2 Sampling point

It is necessary to ensure that the gas concentrations measured are representative of the average conditions inside the waste gas duct. In most cases, a single sampling point situated in the middle of the duct shall be selected. For larger ducts, this point can be situated closer to the sampling port but not too close to avoid any disturbance of the flow or concentration due to influences from the sampling port.

However, when non-homogeneity of the flue gas is suspected, it shall be demonstrated if the flue gas is homogeneous or not. The homogeneity can be demonstrated with a continuous measurement of O_2 or CO_2 using a fixed and a moving probe. Sampling in non-homogeneous flue gases requires following the relevant European Standards or technical specifications proposed by CEN/TC 264.

6.2.3 Assembling the equipment

When using a condensation unit with impingers or absorbers, these shall be filled with water, the volume of which is less than half the content of the absorber.

The water can be replaced by the appropriate absorption solution used in EN 14791, EN 1911-3 or in any other European Standards using water in the preparation of the absorption solution, when the operator wants to collect water vapour with the same sampling train as for SO_2, HCl or any other component. When using an adsorption unit, fill the last impinger or absorber or cartridge with a drying agent.

NOTE This European Standard has only been validated with silica gel even if this commonly used drying agent can absorb carbon dioxide; this error has been accepted.

Assemble the trapping unit, including junctions. Weigh and record the elements of the trapping unit including junctions, with a resolution equal or better than 0,1 g or 2,0 % of the water weight to be measured. Then assemble the whole sampling train.

If applicable, turn on the probe heater and the filter heating system to a controlled temperature of at least 120 °C and 20 °C higher than the (acid) dew point of gases, to prevent water condensation in front of the condenser. Allow time for the temperatures to stabilise.

6.3 Leak test

Perform a leak test on the sampling train. Check the sampling line for leakage according to the following procedure or any other relevant procedure:

— assemble the complete sampler system, including charging the filter housing and absorbers;

— seal the nozzle inlet;

— switch on the pump(s);

— after reaching minimum pressure read the flow rate;

— leak flow rate shall be measured (e.g; by a rotameter) and shall not exceed 2 % of the expected sample gas flow rate.

Perform the leak test at the operating temperature unless this conflicts with safety requirements.

Integrity of the sampling system can be also tested during sampling by continuously measuring the concentration of a suitable stack gas component (e.g. O_2) directly in the stack and downstream the sampling line. Any systematic difference between those concentrations indicates a leak in the system.

6.4 Performing of the sampling

6.4.1 Introduction of the probe in the duct

Immediately after the leak test, insert the sampling probe through the sampling port and place the probe at the chosen representative measurement point within the measurement plane.

Seal the resultant space around the sampling probe and the access hole with an appropriate sealing material such that ambient air is not induced into the duct (nor should any flue gas escape from the duct). In cases where large pressure differences exist between the ambient air and sample gas, care shall be taken that no condensate leaves the absorber.

Control the jacket and filter holder temperatures.

6.4.2 Sampling

Start the sampling pump and adjust the regulating valve to give the desired sample gas-volume flow rate. The selected sample gas flow-rate shall be kept within ± 10 % of the nominal flow rate (see note 2). However, when non-homogeneity is suspected, it shall be demonstrated if the flue gas is homogeneous or not. Sampling in non-homogeneous flue gases shall follow the requirements of relevant Standards or Technical Specifications proposed by CEN/TC 264.

Record the reading on the gas meter V_1 and the time.

Record the reading on the gas meter temperature device T_j and the absolute pressure P_m on the gas meter at least 5 min after starting sampling and at the end of the sampling period.

NOTE For a sampling train as shown in Figure B.1 in Annex B, with gas meter placed downstream pump, absolute pressure on the gas meter is close to atmospheric pressure.

During the whole sampling run, check that the trapping capability of the trapping unit is not exceeded. This can be achieved:

— either by measuring the temperature at the outlet of trapping unit. This temperature shall not be greater than 4 °C; or

— by checking visually the amount of silica gel having faded in the last impinger or cartridge. This amount shall not exceed 50 %.

At the end of the sampling period, switch off the sampling pump, record the time and the reading on the gas meter V_2.

Disassemble the sampling train (adsorption unit). When using a cooling bath, the outside of the trapping unit is thoroughly wiped (e.g. with a dry cloth) before weighing. Next, determine the increase in weight of the whole trapping unit. Record the weighing results.

6.5 Repeatability of the weighing

To determine the repeatability of the weighing in the field in order to take into account the influence of environmental conditions:

— carry out 10 independent weightings of the trapping unit;

— calculate the mean value and the standard deviation.

6.6 Procedure for gas streams saturated with water (droplets present)

When droplets are present, the method produces values for water vapour which are larger than those would be expected for saturated conditions at the temperature of flue gas. Under such conditions the result is consequently biased; therefore water vapour concentration can be determined by using the table of Annex A. This table shows the water vapour concentration for saturated gases for the corresponding average temperature of the flue gas in the sampling plane.

In order to measure the temperature, the probe is inserted at a representative point in the stack. Care shall be taken to wait for the stabilisation of the recorded temperature.

7 Determination of the characteristics of the method: sampling and analysis

7.1 Introduction

When this European Standard is used as the SRM, the user shall demonstrate that the performance characteristics of the method given in Table 1 are better than the minimum performance criteria and that the overall uncertainty calculated by combining values of selected performance characteristics by means of an uncertainty budget is less than ± 20,0 % relative of the measured value.

The values of the selected performance characteristics can be evaluated by means of a laboratory test and a field test implemented by the user, providing that this user is recognised by the competent authority (e.g. through an accreditation system).

7.2 Relevant performance characteristics of the method and performance criteria

The uncertainty of the measured values is influenced by:

— sampling system;

— sampling procedure;

— weighing;

— specific site conditions.

The evaluation of the uncertainty is described in 7.3.

Table 1 gives an overview of the relevant performance characteristics and minimum performance criteria, which shall be determined during the laboratory and field tests.

Table 1 — Minimum performance characteristics of the measurement method

Performance characteristic	Lab. test	Field test	Performance criterion
Volume gas meter:			
— sample volume[a]	X[a]		uncertainty ≤ ± 2,0 % of the volume of sampled gas[b]
— temperature[a]	X[a]		uncertainty ≤ ± 2,5 K[b]
— absolute pressure[a]	X[a]		uncertainty ≤ ± 1,0 % of the absolute pressure[b]
Leak in the sampling line		X	≤ 2,0 % of the nominal flow rate
Weighing of collected water:			
— uncertainty associated to the balance[c]	X	X	
— repeatability in the field of weighing[d]		X	

[a] The uncertainty of sampling volume gas is a combination of uncertainties due to calibration, resolution, reading and drift. The uncertainty related to the temperature and absolute pressure of the gas meter is a combination of the uncertainties due to calibration, resolution, reading, drift and repeatability. When a barometer is used see chapter 5.9.

[b] Performance criteria correspond to the uncertainty of calibration.

[c] The uncertainty contributions of the balance can be for example: linearity of balance, resolution, reading and calibration tare. See chapter 5.10.

[d] When weighing is carried out in the field, variations of ambient conditions can be an uncertainty source, for example temperature variations, air currents and vibrations.

7.3 Establishment of the uncertainty budget

An uncertainty budget shall be established to evaluate whether or not the method fulfils the requirements for a maximum allowable overall uncertainty. This uncertainty budget shall be drawn up according to the procedures described in ENV 13005, taking into account at least all the relevant characteristics given in Table 1.

The overall uncertainty for this method used as a reference shall be lower than ± 20,0 % of the measured value, before correction at the reference concentration of O_2.

The principle of calculation of the overall uncertainty is based on the law on propagation of uncertainty laid down in the ENV 13005:

— determine the standard uncertainties attached to the performance characteristics to be included in the calculation of the uncertainty budget by means of laboratory or field tests, and according to ENV 13005;

— calculate the uncertainty budget by combining all the standard uncertainties according to ENV 13005, and taking the variations range of the influence quantities of the specific site conditions into account;

— all the components of standard uncertainties that are less than 5 % relative of the maximum standard uncertainty can be neglected;

— calculate the overall uncertainty at the measured value.

An example of the evaluation of the overall uncertainty is given in Annex C.

7.4 Equivalency with an alternative method

In order to show that an alternative method is equivalent to the reference method, follow the procedures described in the CEN/TS 14793.

The maximum allowable standard deviation of repeatability expressed in % for this reference method is:

$$s_{\mathrm{r\,limit}}(C) = 0{,}0218C + 0{,}247 \quad (1)$$

where

C is the concentration in %.

The standard deviation of reproducibility expressed in % for this reference method is:

$$s_{\mathrm{R}}(C) = 0{,}03C + 0{,}37 \quad (2)$$

where

C is the concentration in %.

8 Evaluation of the method in the field

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different measuring teams originating from ten European Union member states.

The characteristics of installations, the conditions during field tests and the values of repeatability and reproducibility in the field are given in Annex D.

9 Water vapour determination

Calculate the mean value temperature T_m at the gas meter:

$$T_m = (T_1 + T_2 + ... + T_n)/n(K) \tag{3}$$

where

n is the number of the temperature T_j readings got during the test.

Calculate the dry gas volume sampled from the gas meter, at standard conditions, in cubic meter $V_{m(std)}$:

if a dry-gas volume meter is used $V_{m(std)} = (V_2 - V_1) \times \frac{T_{std}}{T_m} \times \frac{P_m}{P_{std}}$ (4)

if a wet-gas volume meter is used $V_{m(std)} = (V_2 - V_1) \times \frac{T_{std}}{T_m} \times \frac{P_m - P_{sat(T_m)}}{P_{std}}$ (5)

where

$V_{m(std)}$ is the dry gas volume measured, corrected to standard conditions, in m3;

V_2-V_1 is the sampled gas volume, at actual conditions of temperature, pressure and humidity, in m3;

Tm is the mean temperature of the sampled gas at the gas-meter, in K;

Tstd is the standard temperature, 273 K;

Pm is the absolute pressure at the gas meter, in kPa;

$P_{sat(Tm)}$ is the saturation vapour pressure of water at the temperature of the gas meter, in kPa;

P_{std} is the standard pressure, 101,3 kPa.

The water vapour content in grams per cubic meter in standardised conditions of temperature and pressure and on dry basis is:

$V_{wc(g/m3)} = \frac{m_{wc}}{V_{m(std)}}$ in g/m^3 (6)

where

$V_{wc(g/m3)}$ is the water vapour content on dry basis, in g/m^3;

$V_{m(std)}$ is the dry gas volume measured, corrected to standard conditions, in m^3;

m_{wc} is the mass of water vapour trapped in the trapping unit, in g.

The water vapour content in % volume on wet basis is:

$$V_{wc(\%)}=\frac{\dfrac{m_{wc}\times V_{mol(std)}}{M_w}}{\dfrac{m_{wc}\times V_{mol(std)}}{M_w}+V_{m(std)}}\times 100 \tag{7}$$

where

$V_{wc(w,\%)}$ is the water vapour content on wet basis, in % volume (m^3 of water vapour in m^3 of wet gas);

$V_{m(std)}$ is the dry gas volume measured, corrected to standard conditions, in m^3;

m_{wc} is the mass of water vapour collected in the trapping unit, in g;

M_w is the molecular weight of water, 18,01534 g/mol rounded to 18 g/mol;

$V_{mol(std)}$ is the molar volume at standard conditions, in m^3/mol at P_{std} and T_{std}.

Concentrations are generally expressed at a reference concentration of O_2 defined in the European Directives:

— 11 % for incineration of waste in waste incinerators;

— 10 % for co-incineration of waste in cement kilns;

— 3 % for combustion of gas or liquid fuels;

— 6 % for combustion of solid fuels;

— 15 % for gas turbines.

The corrected concentration of measurand C_{corr} is calculated using the followed equation:

$$C_{corr}=\frac{21-0_{2,ref}}{21-0_{2,mes}}\times C_{actual} \tag{8}$$

where

C_{corr} is the corrected concentration of measurand;

C_{actua} is the measured concentration of measurand at actual O_2 concentration;

$O_{2,mes}$ is the measured mean dry oxygen content during the sampling time;

$O_{2,ref}$ is the oxygen reference concentration.

10 Report

The test report shall include the following information:

a) reference to this European Standard;

b) description of the purpose of tests;

c) information about the personnel involved in the measurement;

d) description of plant and process;

e) description of the operating conditions of the plant process;

f) identification of the sampling plane;

g) description of the location of the sampling point(s) in the sampling plane;

h) changes in the plant operations during sampling, for example burner changes;

i) characteristics of sampling equipment;

j) number of tests;

k) for each test : date, time, duration and identification of samples(s);

l) test results : sampling volume, concentrations;

m) trapping efficiency;

n) actual uncertainty budget established according to EN ISO 14956 or ENV 13005;

o) any deviations from this European Standard.

Annex A
(normative)

Determination of water vapour concentration for water saturated gas, at P_{std} = 101,325 kPa

Temperature T of gas in duct	Vapour pressure	Water vapour concentration	Water vapour concentration at T and P_{std}	Water vapour concentration at T and P_{std}	Water vapour concentration
°C	k Pa	g/kg wet air	g/m^3 of wet gas	g/m^3 of dry gas	% volume of wet gas
0	0,614	3,78	4,87	4,90	0,61
1	0,660	4,06	5,22	5,25	0,65
2	0,709	4,36	5,59	5,63	0,70
3	0,761	4,69	5,98	6,02	0,75
4	0,817	5,03	6,39	6,44	0,81
5	0,876	5,40	6,83	6,89	0,86
6	0,939	5,79	7,30	7,36	0,93
7	1,006	6,20	7,79	7,87	0,99
8	1,078	6,64	8,31	8,40	1,06
9	1,153	7,11	8,86	8,96	1,14
10	1,234	7,61	9,44	9,56	1,22
11	1,319	8,13	10,06	10,19	1,30
12	1,409	8,69	10,71	10,86	1,39
13	1,505	9,29	11,40	11,57	1,48
14	1,606	9,92	12,12	12,32	1,58
15	1,713	10,58	12,89	13,11	1,69
16	1,827	11,29	13,70	13,95	1,80
17	1,947	12,04	14,55	14,83	1,92
18	2,074	12,83	15,44	15,76	2,05
19	2,208	13,67	16,38	16,75	2,18
20	2,350	14,55	17,38	17,79	2,32
21	2,499	15,49	18,42	18,89	2,47
22	2,657	16,47	19,52	20,04	2,62
23	2,824	17,52	20,67	21,26	2,79
24	2,999	18,62	21,88	22,55	2,96
25	3,185	19,78	23,15	23,91	3,14
26	3,379	21,01	24,49	25,34	3,34
27	3,585	22,30	25,89	26,84	3,54

Temperature T of gas in duct	Vapour pressure	Water vapour concentration	Water vapour concentration at T and P_{std}	Water vapour concentration at T and P_{std}	Water vapour concentration
°C	k Pa	g/kg wet air	g/m^3 of wet gas	g/m^3 of dry gas	% volume of wet gas
28	3,801	23,67	27,36	28,43	3,75
29	4,028	25,10	28,90	30,10	3,98
30	4,267	26,62	30,52	31,86	4,21
31	4,519	28,21	32,21	33,71	4,46
32	4,783	29,89	33,98	35,66	4,72
33	5,060	31,66	35,83	37,71	4,99
34	5,351	33,52	37,77	39,87	5,28
35	5,657	35,47	39,80	42,15	5,58
36	5,978	37,53	41,92	44,54	5,90
37	6,314	39,69	44,13	47,07	6,23
38	6,667	41,97	46,45	49,72	6,58
39	7,036	44,35	48,86	52,51	6,94
40	7,423	46,86	51,39	55,45	7,33
41	7,828	49,50	54,02	58,54	7,73
42	8,253	52,27	56,77	61,80	8,14
43	8,697	55,17	59,63	65,23	8,58
44	9,161	58,23	62,62	68,84	9,04
45	9,647	61,43	65,73	72,65	9,52
46	10,155	64,79	68,97	76,66	10,02
47	10,685	68,31	72,35	80,88	10,55
48	11,239	72,01	75,87	85,33	11,09
49	11,818	75,89	79,52	90,02	11,66
50	12,422	79,96	83,33	94,97	12,26
51	13,052	84,22	87,29	100,19	12,88
52	13,710	88,69	91,40	105,71	13,53
53	14,395	93,38	95,68	111,52	14,21
54	15,110	98,29	100,12	117,67	14,91
55	15,855	103,44	104,74	124,17	15,65
56	16,631	108,84	109,53	131,04	16,41
57	17,439	114,50	114,51	138,31	17,21
58	18,281	120,43	119,67	146,01	18,04
59	19,157	126,64	125,03	154,18	18,91
60	20,068	133,16	130,58	162,83	19,81
61	21,016	139,98	136,34	172,02	20,74
62	22,002	147,14	142,31	181,78	21,71

Temperature T of gas in duct	Vapour pressure	Water vapour concentration	Water vapour concentration at T and P_{std}	Water vapour concentration at T and P_{std}	Water vapour concentration
°C	k Pa	g/kg wet air	g/m^3 of wet gas	g/m^3 of dry gas	% volume of wet gas
63	23,027	154,63	148,50	192,17	22,73
64	24,092	162,49	154,90	203,22	23,78
65	25,199	170,73	161,54	215,01	24,87
66	26,348	179,37	168,41	227,59	26,00
67	27,541	188,42	175,52	241,03	27,18
68	28,780	197,91	182,87	255,42	28,40
69	30,065	207,87	190,48	270,85	29,67
70	31,399	218,31	198,35	287,42	30,99
71	32,781	229,27	206,48	305,24	32,35
72	34,215	240,76	214,89	324,45	33,77
73	35,701	252,83	223,58	345,21	35,23
74	37,241	265,49	232,55	367,69	36,75
75	38,837	278,79	241,81	392,10	38,33
76	40,489	292,76	251,38	418,68	39,96
77	42,199	307,44	261,25	447,71	41,65
78	43,970	322,87	271,43	479,52	43,39
79	45,801	339,09	281,94	514,51	45,20
80	47,696	356,16	292,77	553,16	47,07
81	49,656	374,12	303,94	596,04	49,01
82	51,682	393,03	315,45	643,86	51,01
83	53,776	412,95	327,31	697,48	53,07
84	55,940	433,95	339,52	758,00	55,21
85	58,175	456,09	352,10	826,80	57,41
86	60,483	479,46	365,05	905,65	59,69
87	62,865	504,13	378,38	996,87	62,04
88	65,325	530,21	392,09	1103,56	64,47
89	67,862	557,79	406,19	1229,95	66,97
90	70,479	586,97	420,70	1381,95	69,56
91	73,178	617,89	435,61	1568,12	72,22
92	75,960	650,67	450,93	1801,33	74,97
93	78,827	685,46	466,67	2101,78	77,80
94	81,781	722,42	482,84	2503,25	80,71
95	84,823	761,74	499,44	3066,66	83,71
96	87,955	803,61	516,48	3914,24	86,81
97	91,179	848,25	533,96	5332,65	89,99

Temperature T of gas in duct	**Vapour pressure**	**Water vapour concentration**	**Water vapour concentration at T and P_{std}**	**Water vapour concentration at T and P_{std}**	**Water vapour concentration**
°C	k Pa	g/kg wet air	g/m^3 of wet gas	g/m^3 of dry gas	% volume of wet gas
98	94,496	895,91	551,90	8189,39	93,26
99	97,909	946,88	570,29	16914,52	96,63
100	-	-	-	-	-

Pressure correction:

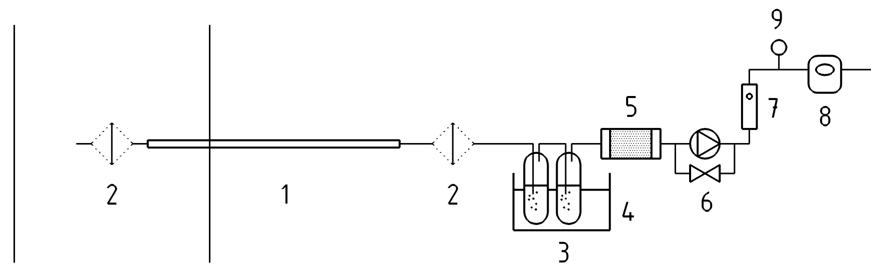
$$V_{\mathrm{wc(T,P)}}(\mathrm{g/m^3}) = V_{\mathrm{wc(T,Pstd)}}(\mathrm{g/m^3}) \times \frac{P}{P_{\mathrm{std}}} \qquad \text{(A.1)}$$

where

P is the absolute gas pressure of gas in duct.

Annex B
(informative)

Type of sampling equipments

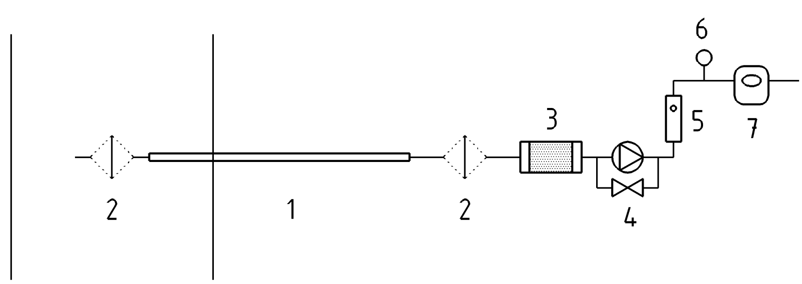


Key

1 Heated probe
2 Heated filter (in-stack or out-stack)
3 Impingers or absorbers
4 Cooling bath (optional)
5 Adsorption cartridge
6 Pump
7 Gas flow meter
8 Gas volume meter
9 Temperature and pressure measurement

NOTE If the pressure drop in the gas meter is lower than 100 Pa, the absolute pressure is equal to the atmospheric pressure.

Figure B.1 — Trapping unit with two stages : condensation and adsorption



Key

1 Heated probe
2 Heated filter (in-stack or out-stack)
3 Adsorption cartridge
4 Pump
5 Gas flow meter
6 Temperature and pressure measurement
7 Gas volume meter

Figure B.2 — Trapping unit with one adsorption stage

Annex C
(informative)

Example of assessment of compliance of reference method for water vapour with requirements on emission measurements

C.1 General

This informative Annex gives an example of calculation of the overall uncertainty established with configuration 1.

C.2 Process of uncertainty estimation

The procedure for calculating measurement uncertainty as follows is based on the law on propagation of uncertainty laid down in EN ISO 14956 or ENV 13005. The calculation procedure presents different steps:

C.2.1 Determination of the model equation

Define the measurand and all the parameters that influence the result of the measurement. These parameters, called “input quantities” shall be clearly defined.

Identify all sources of uncertainty contributing to any of the input quantities or to the measurand directly.

Then the model equation, that is to say the relationship between the measurand and the influence quantities, shall be established, if possible in mathematical equation form.

C.2.2 Quantification of uncertainty components

Each uncertainty source is estimated to obtain its contribution to the overall uncertainty.

Use available performance characteristics of the measurement system, data from the dispersion of repeated measurements, data provided in calibration certificates.

Convert all uncertainty components (e.g. performance characteristics) to standard uncertainties of input and influence quantities.

C.2.3 Calculation of the combined uncertainty

Then the combined uncertainty u_{c} is calculated by combining standard uncertainties, by applying the "law of propagation of uncertainty".

In general, the uncertainty associated to a concentration is expressed in overall uncertainty form. The overall uncertainty U_{c} corresponds to the expanded combined uncertainty, obtained by multiplying by a coverage factor "k" : $U_{\mathrm{c}}=k\times u_{\mathrm{c}}$. The value of the coverage factor k is chosen on the basis of the level of confidence required. In most cases, k is taken equal to 2, for a level of confidence of approximately 95 %.

The following Tables give one example of:

— specific conditions of the site (Table C.1), that is to say the values and the variation range of the influence parameters;

— performance characteristics of the method (Table C.3) related to the parameters which can have an influence on the response of the analyser: the performance characteristics of the analyser and the effect of influence parameters (e.g. chemical interferents, and environmental conditions such as ambient temperature, voltage and pressure).

C.3 Specific conditions in the field

Measurements are carried out in a duct where the mass flow is homogenous.

Table C.1 — Measurement conditions

Specific conditions	**Value/range**
Concentration	17,9 % volume on wet basis
Water collected in the trapping unit[a] : weight of water collected in the condensation unit weight of water trapped in adsorption unit	 17,2 g 2,0 g
Volume of gas sampled in 30 min	0,120 m^3
Nominal flow rate Q	240 l/h
Mean temperature (in Kelvin) at the gas meter[b]	296 K
Mean absolute pressure at the gas meter[c]	100 281 Pa

[a] A calibrated balance is used to determine water vapour collected in the condensation unit and in the adsorption unit. The procedure consists in weighing the trapping unit before and after sampling.

Repeatability of the weighing in the field has been estimated by 10 independent weighing of a trapping unit: the standard deviation is equal to 0,78 g.

[b] The mean temperature is calculated from data recording of temperature measurements (1 measurement / 30 s ➔ 60 measurements in 30 min):

— The standard deviation of measurements is equal to 3,1 K.

— Drift: no drift between 2 calibrations.

[c] The mean absolute pressure is calculated from 5 measurements of the static relative pressure at the gas meter and 1 measurement of the atmospheric pressure during the sampling period.

Table C.2 — Measurement values of relative pressure

Measurement	1	2	3	4	5	Mean	Standard deviation of measured values (Pa)
Relative pressure at gas meter P_{rel} (Pa)	70,5	68,0	65,0	69,0	70,5	68,6	2,27

Barometric pressure:100 212 Pa.

Mean absolute pressure:100 281 Pa.

C.4 Performance characteristics of the method

Table C.3 — Performance characteristics of the method

Performance characteristics for the reference method	Performance criteria	Results laboratory or field tests
Weighing of water vapour collected: — Adjustment with tare — Reading — Standard deviation of repeatability of the weighing in the field		 100 g ± 0,01 g 0,02 g 0,5 g
Gas volume meter: Overall uncertainty Standard deviation of repeatability Drift between 2 adjustments Reading	 ≤ ± 2,0 % of the measured value	 ± 2 % of the measured value 0,3 % of the measured value 2 % of the measured value 0,002 m^3
Temperature at the gas meter: Overall uncertainty — Standard deviation of repeatability — Resolution	 ≤ ± 2,5 K	 ± 0,5 K 0,02 K 0,01 K
Absolute pressure at the gas meter: Overall uncertainty — Manometer at the gas meter — Atmospheric pressure	 ≤ ± 1,0 % of the measured value	 ± 1,5 Pa for manometer ± 170 Pa for barometer
Relative pressure at gas volume meter: — Std deviation of repeatability — Resolution Atmospheric pressure : — Drift between 2 adjust. — Reading		 0,3 Pa 0,5 Pa 60 Pa 10 Pa
Leakage in the sampling line	≤ 2,0 % of the nominal flow rate	1,5 % of the nominal flow rate

C.5 Calculation of standard uncertainty of the concentration

In this clause, the model equations, as well as the calculations of the partial uncertainties are related to the measured concentration, taking into account the collection efficiency.

C.5.1 Model equation and application of the rule of the uncertainty propagation

C.5.1.1 Concentration of water vapour

The concentration of water vapour measured is calculated by equations (C.1) and (C.2):

$$V_{\text{wc,meas}} = \frac{\dfrac{m_{\text{wc}} \times V_{\text{mol(std)}}}{M_{\text{w}}}}{\dfrac{m_{\text{wc}} \times V_{\text{mol(std)}}}{M_{\text{w}}} + V_{\text{m(std)}}} 100 = \frac{\dfrac{(m_{\text{wc(cs)}} + m_{\text{wc(as)}}) \times V_{\text{mol(std)}}}{M_w}}{\dfrac{(m_{\text{wc(cs)}} + m_{\text{wc(as)}}) \times V_{\text{mol(std)}}}{M_{\text{w}}} + V_{\text{m(std)}}} \times 100 \tag{C.1}$$

$$V_{\text{m(std)}} = V_{\text{m}} \times \frac{T_{\text{std}}}{T_{\text{m}}} \times \frac{P_{\text{m}}}{P_{\text{std}}} = V_{\text{m}} \times \frac{T_{\text{std}}}{T_{\text{m}}} \times \frac{P_{\text{rel}} + P_{\text{atm}}}{P_{\text{std}}} \tag{C.2}$$

where

$V_{\text{wc,meas}}$ is the water vapour content on wet basis, in % volume (m^3 of water vapour in m^3 of wet gas);

m_{wc} is the total mass of water vapour collected in the trapping unit, in g;

$m_{\text{wc(cs)}}$ is the mass of water vapour collected in the condensation stage, in g;

$m_{\text{wc(as)}}$ is the mass of water vapour collected in the adsorption stage, in g;

M_{w} is the molecular weight of water, 18 g/mol;

$V_{\text{mol(std)}}$ is the molar volume at standard conditions, in m^3/mol;

$V_{\text{mol(std)}} =$ $22{,}4.10^{-3}$ m^3/mol at 273 K and 101,3 kPa;

$V_{\text{m(std)}}$ is the sampled dry gas volume from the gas meter, at standard conditions, in m^3;

V_{m} is the sampled gas volume (in cubic meter) calculated by difference between the gas volume at the end and at the beginning of the sampling period. The value at the beginning of the sampling period corresponds to a reading of an indicator; the value at the end of the sampling period corresponds to the reading of a measured value;

T_{m} is the mean temperature of the sampled gas at the gas-meter, in K;

T_{std} is the standard temperature, 273 K;

$P_{\text{m}} = P_{\text{rel}} + P_{\text{atm}}$ is the absolute pressure (in kilopascals) at the gas volume meter; P_m is equal to the sum of relative pressure measured at the gas volume meter P_{rel} plus atmospheric pressure P_{atm};

P_{std} is the standard pressure, 101,3 kPa.

C.5.1.2 Effect of the collection efficiency

The efficiency of the sampling system can be checked by placing an additional absorber unit after the "normal" trapping unit. The weight of the water collected in it is determined with the same weighing procedure and with the same uncertainty than the water collected in the "normal" trapping unit. The efficiency of the sampling system can also be checked by calculating the ratio between the measured water vapour concentration and the remaining water vapour concentration at the exit of the adsorption stage. When the result has been corrected for the remaining water vapour at the outlet of the adsorption stage, then it is possible to consider that $\frac{100}{\varepsilon}=1$.

The collection efficiency is calculated according to the equation D.3:

$$\varepsilon=\frac{m_{\text{wc(cs)}}+m_{\text{wc(as)}}}{(m_{\text{wc(cs)}}+m_{\text{wc(as)}})+m_{\text{wc(add)}}}\times 100 \tag{C.3}$$

where

ε collection efficiency in %;

$m_{\text{wc(cs)}}$ is the mass of water vapour collected in the condensation stage, in g;

$m_{\text{wc(as)}}$ is the mass of water vapour collected in the adsorption stage, in g;

$m_{\text{wc(cs)}}+m_{\text{wc(as)}}$ is the weight of the water vapour collected in the "normal" trapping unit;

$m_{\text{wc(add)}}$ is the weight of the water vapour collected in the additional absorber.

The mass of water vapour in sampling gas from the duct is then equal to: $\frac{(m_{\text{wc(cs)}}+m_{\text{wc(as)}})\times 100}{\varepsilon}$

C.5.1.3 Calculation of the combined uncertainty of V_{wc}, taking into account the collection efficiency

$$V_{\text{wc}}=\frac{\frac{m_{\text{wc}}\times V_{\text{mol(std)}}}{M_{\text{w}}}}{\frac{m_{\text{wc}}\times V_{\text{mol(std)}}}{M_{\text{w}}}+V_{\text{m(std)}}}100=\frac{\frac{(m_{\text{wc(cs)}}+m_{\text{wc(as)}})\times 100\times V_{\text{mol(std)}}}{\varepsilon\times M_{\text{w}}}}{\frac{(m_{\text{wc(cs)}})+m_{\text{wc(as)}})\times 100\times V_{\text{mol(std)}}}{\varepsilon\times M_{\text{w}}}+V_{\text{m(std)}}}\times 100$$

Hypothesis : we consider that uncertainties of $V_{\text{mol(std)}}$ and M_{w} are negligible.

$$u^2(V_{\text{wc}})=\left(\frac{\partial V_{\text{wc}}}{\partial m_{\text{wc(cs)}}}\right)^2\times u^2(m_{\text{wc(cs)}})+\left(\frac{\partial V_{\text{wc}}}{\partial m_{\text{wc(as)}}}\right)^2\times u^2(m_{\text{wc(as)}})$$
$$+\left(\frac{\partial V_{\text{wc}}}{\partial \varepsilon}\right)^2\times u^2(\varepsilon)+\left(\frac{\partial V_{\text{wc}}}{\partial V_{\text{m(std)}}}\right)^2\times u^2(V_{\text{m(std)}}) \tag{C.4}$$

C.5.1.4 Calculation of sensitivity coefficients:

$$\frac{\partial V_{\text{wc}}}{\partial m_{\text{wc(cs)}}}=\frac{V_{\text{wc}}\times v_{\text{m(std)}}}{\frac{(m_{\text{wc(cs)}}+m_{\text{wc(as)}})\times 100\times V_{\text{mol}}}{\varepsilon\times M_{\text{w}}}+V_{\text{m(std)}}}\times\frac{1}{(m_{\text{wc(cs)}}+m_{\text{wc(as)}})} \tag{C.5}$$

$$\frac{\partial V_{wc}}{\partial m_{wc(as)}}=\frac{V_{wc}\times v_{m(std)}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol}}{\varepsilon\times M_w}+V_{m(std)}}\times\frac{1}{(m_{wc(cs)}+m_{wc(as)})} \quad (C.6)$$

$$\frac{\partial V_{wc}}{\partial \varepsilon}=\frac{-V_{wc}\times V_{m(std)}\times\frac{1}{\varepsilon}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol}}{\varepsilon\times M_w}+V_{m(std)}} \quad (C.7)$$

$$\frac{\partial V_{wc}}{\partial V_{m(std)}}=\frac{-V_{wc}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol}}{\varepsilon\times M_w}+V_{m(std)}} \quad (C.8)$$

(C.4) is equivalent to:

$$\frac{u^2(V_{wc})}{(V_{wc})^2}=\left(\frac{v_{m(std)}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol}}{\varepsilon\times M_w}+V_{m(std)}}\right)^2\times\left(\left(\frac{u(m_{wc(cs)})}{(m_{wc(cs)}+m_{wc(as)})}\right)^2+\left(\frac{u(m_{wc(as)})}{(m_{wc(cs)}+m_{wc(as)})}\right)^2\right.$$

$$\left.+\left(\frac{u(\varepsilon)}{\varepsilon}\right)^2+\left(\frac{u(V_{m(std)})}{V_{m(std)}}\right)^2\right) \quad (C.9)$$

C.5.1.5 Calculation of the standard uncertainty of ε:

$$\varepsilon=\frac{m_{wc(cs)}+m_{wc(as)}}{(m_{wc(cs)}+m_{wc(as)})+m_{wc(add)}}\times 100 \quad (C.3)$$

$$m_{wc(add)}=\frac{(m_{wc(cs)}+m_{wc(as)})\times(100-\varepsilon)}{\varepsilon} \quad (C.10)$$

$$u^2(\varepsilon)=\left(\frac{\partial\varepsilon}{\partial m_{wc(add)}}\right)^2\times u^2(m_{wc(add)})+\left(\frac{\partial\varepsilon}{\partial m_{wc(cs)}}\right)^2\times u^2(m_{wc(cs)})+\left(\frac{\partial\varepsilon}{\partial m_{wc(as)}}\right)^2\times u^2(m_{wc(as)}) \quad (C.11)$$

$$\frac{\partial\varepsilon}{\partial m_{wc(cs)}}=\frac{100\times m_{wc(add)}}{(m_{wc(cs)}+m_{wc(as)}+m_{wc(add)})^2}=\frac{100-\varepsilon}{100}\times\frac{\varepsilon}{(m_{wc(cs)}+m_{wc(as)})} \quad (C.12)$$

$$\frac{\partial\varepsilon}{\partial m_{wc(as)}}=\frac{100\times m_{wc(add)}}{(m_{wc(cs)}+m_{wc(as)}+m_{wc(add)})^2}=\frac{100-\varepsilon}{100}\times\frac{\varepsilon}{(m_{wc(cs)}+m_{wc(as)})} \quad (C.13)$$

$$\frac{\partial\varepsilon}{\partial m_{wc(add)}}=\frac{-100\times(m_{wc(cs)}+m_{wc(as)})}{(m_{wc(cs)}+m_{wc(as)}+m_{wc(add)})^2}=\frac{-(100-\varepsilon)}{100}\times\frac{\varepsilon}{m_{wc(add)}} \quad (C.14)$$

(C.11) is equivalent to (C.15):

$$\frac{u^2(\varepsilon)}{\varepsilon^2}=\left(\frac{100-\varepsilon}{100}\right)^2\times\left(\frac{u^2(m_{wc(cs)})}{(m_{wc(cs)}+m_{wc(as)})^2}+\frac{u^2(m_{wc(as)})}{(m_{wc(cs)}+m_{wc(as)})^2}+\frac{u^2(m_{wc(add)})}{(m_{wc(add)})^2}\right) \quad (C.15)$$

C.5.1.6 Calculation of the standard uncertainty of $V_{m(std)}$

$$V_{m(std)} = V_m \times \frac{T_{std}}{T_m} \times \frac{P_m}{P_{std}} = V_m \times \frac{T_{std}}{T_m} \times \frac{P_{rel}+P_{atm}}{P_{std}} \quad \text{(C.2)}$$

Hypothesis: we consider that uncertainties of T_{std} and P_{std} are negligible.

$$u^2(V_{m(std)}) = \left(\frac{\partial V_{m(std)}}{\partial V_m}\right)^2 \times u^2(V_m) + \left(\frac{\partial V_{m(std)}}{\partial T_m}\right)^2 \times u^2(T_m) + \left(\frac{\partial V_{m(std)}}{\partial P_{rel}}\right)^2 \times u^2(P_{rel}) + \left(\frac{\partial V_{m(std)}}{\partial P_{atm}}\right)^2 \times u^2(P_{atm}) \quad \text{(C.16)}$$

Calculation of sensitivity coefficients:

$$\frac{\partial V_{m(std)}}{\partial V_m} = \frac{T_{std}}{T_m} \times \frac{P_m}{P_{std}} = \frac{V_{m(std)}}{V_m} \quad \text{(C.17)}$$

$$\frac{\partial V_{m(std)}}{\partial V_m} = V_m \times T_{std} \times \frac{P_m}{P_{std}} \times \frac{1}{T_m^2} = \frac{V_{m(std)}}{T_m} \quad \text{(C.18)}$$

$$\frac{\partial V_{m(std)}}{\partial P_{rel}} = V_m \times \frac{T_{std}}{T_m} \times \frac{1}{P_{std}} = \frac{V_{m(std)}}{P_{rel}+P_{atm}} = \frac{V_{m(std)}}{P_m} \quad \text{(C.19)}$$

$$\frac{\partial V_{m(std)}}{\partial P_{atm}} = V_m \times \frac{T_{std}}{T_m} \times \frac{1}{P_{std}} = \frac{V_{m(std)}}{P_{rel}+P_{atm}} = \frac{V_{m(std)}}{P_m} \quad \text{(C.20)}$$

(C.16) is equivalent to (C.21):

$$\frac{u^2(V_{m(std)})}{\left(V_{m(std)}\right)^2} = \frac{u^2(V_m)}{(V_m)^2} + \frac{u^2(T_m)}{(T_m)^2} + \frac{u^2(P_{rel})}{(P_{rel}+P_{atm})^2} + \frac{u^2(P_{atm})}{(P_{rel}+P_{atm})^2} \quad \text{(C.21)}$$

C.5.1.7 Calculation of the combined uncertainty

(C.9) is equivalent to (C.22):

$$\frac{u^2(V_{wc})}{(V_{wc})^2}=\left(\frac{v_{m(std)}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol(std)}}{\varepsilon\times M_w}+V_{m(std)}}\right)^2\times\left(\left(\frac{u(m_{wc(cs)})}{(m_{wc(cs)}+m_{wc(as)})}\right)^2+\left(\frac{u(m_{wc(as)})}{(m_{wc(cs)}+m_{wc(as)})}\right)^2\right.$$
$$+\left(\frac{100-\varepsilon}{100}\right)^2\left(\left(\frac{u(m_{wc(cs)})}{m_{wc(cs)}+m_{wc(as)}}\right)^2+\left(\frac{u(m_{wc(as)})}{m_{wc(cs)}+m_{wc(as)}}\right)^2+\left(\frac{u(m_{wc(add)})}{m_{wc(add)}}\right)^2\right)+\left(\frac{u(V_m)}{V_m}\right)^2+\left(\frac{u(T_m)}{T_m}\right)^2$$
$$\left.+\left(\frac{u(P_{rel})}{P_m}\right)^2+\left(\frac{u(P_{atm})}{P_m}\right)^2\right) \quad \text{(C.22)}$$

C.5.2 Results of standard uncertainties calculations

$m_{wc(cs)}$, $m_{wc(as)}$ and $m_{wc(add)}$ are determined by weighing the trapping units before and after sampling:

$m_{wc(cs)}=m_{wc(cs)final}-m_{wc(cs)initial}$;

$m_{wc(as)}=m_{wc(as)final}-m_{wc(as)initial}$;

$m_{wc(add)}=m_{wc(add)final}-m_{wc(add)initial}$.

Uncertainties associated with the weighing before sampling are equals to the ones after sampling. As the weighing of the water vapour collected in each unit is equal to a difference of 2 weightings, systematic errors cancel each other out.

$$u(m_{wc(cs)})=u(m_{wc(as)})=u(m_{wc(add)})=2\times u(m)$$

Where $u(m)$ depends on the uncertainty of the tare, the uncertainty associated to the reading and the repeatability of the weighing.

Table C.4 — Results of uncertainty calculations

Performance characteristics	Standard uncertainty	Value of standard uncertainty at limit value $u(x_i)$	Relative standard uncertainty
Weighing of water vapour collected	$u(m_{wc(cs)})$ $u(m_{wc(as)})$ $u(m_{wc(add)})$	$u^2(m_{wc(cs)})=u^2(m_{wc(as)})$ $=u^2(m_{wc(add)})$ $=2\begin{pmatrix} u^2(\text{adj,m})+u^2(\text{read,m}) \\ +u^2(\text{rep,m}) \end{pmatrix}$	
- adjustment with the tare : $u(adj,m)$		$\frac{0{,}01}{2}=0{,}005$ g	
- reading : $u(read,m)$		$\frac{0{,}02}{2\sqrt{3}}=0{,}006$ g	
- standard deviation of repeatability of the weighing : $u(rep,m)$		0,5 g	
	$u(m_{wc(cs)})$ $=u(m_{wc(as)})$ $=u(m_{wc(add)})$	0,71 g	$\frac{u(m_{wc(cs)})}{m_{wc(cs)}+m_{wc(as)}}$ $=\frac{u(m_{wc(as)})}{m_{wc(cs)}+m_{wc(as)}}=3{,}7.10^{-2}$ $\frac{u(m_{wc(add)})}{m_{wc(add)}}=1{,}8$
Volume of sampled gas	$u(V)$	$u^2(V_m)=u^2(cal,V_m)+u^2(rep,V_m)$ $+u^2(dr,V_m)+2u^2(read,V_m)$	
— calibration : $u(cal,V_m)$		$\frac{2/100\text{x}0{,}120}{2}=0{,}0012$ m^3	
— drift : $u(dr,V_m)$		$\frac{2/100\times 0{,}120}{\sqrt{3}}=0{,}0014$ m^3	
— repeatability std deviation $u(rep,V_m)$		$0{,}3/100\times 0{,}120=3{,}6.10^{-4}$	10^{-2}
— reading : $u(read,V_m)$		$\frac{0{,}002}{2\sqrt{3}}=0{,}0006$ m^3	
	$u(V_m)$	0,002 m^3	$\frac{u(V)}{V_m}=\frac{0{,}002}{0{,}120}=1{,}7.10^{-2}$
Temperature at the gas meter	$u(T_m)$	$u^2(T_m)=u^2(cal,T_m)+u^2(res,T_m)$ $+u^2(rep,T_m)+u^2(mean,T_m)$	
— calibration : $u(cal,T_m)$		$\frac{0{,}5}{2}=0{,}25$ K	

(continued)

Table C.4 *(end)*

Performance characteristic	Standard uncertainty	Value of standard uncertainty at limit value $u(x_i)$	Relative standard uncertainty
— repeatability standard deviation : $u(rep,T_m)$		0,02 K	
— resolution : $u(res,T_m)$		$\frac{0{,}01}{2\sqrt{3}}=0{,}003$ K	
— standard deviation of the mean : $u(mean,T_m)$		$\frac{3{,}1}{\sqrt{60}}=0{,}40$	
	$u(T_m)$	0,47 K	$\frac{u(T_m)}{T_m}=\frac{0{,}47}{296}=1{,}6.10^{-3}$
Relative pressure at the gas meter	$u(P_{rel})$	$u^2(P_{rel})=u^2(cal,P_{rel})+u^2(rep,P_{rel})$ $+u^2(res,P_{rel})+u^2(mean,P_{rel})$	
— calibration of the manometer : $u(cal,P_{rel})$		$\frac{1{,}5}{2}=0{,}75$ Pa	
— repeatability standard deviation : $u(rep, P_{rel})$		0,3 Pa	
— resolution of manometer : $u(res,P_{rel})$		$\frac{0{,}5}{2\sqrt{3}}=0{,}15$ Pa	
— standard uncertainty associated to the mean : $u(mean,P_{rel})$		$\frac{2{,}28}{\sqrt{5}}=1{,}02$ Pa	
	$u(P_{rel})$	1,31 Pa	$\frac{u(P_{rel})}{P_m}=\frac{1{,}28}{100281}=1{,}3.10^{-5}$
Atmospheric pressure	$U(P_{atm})$	$u^2(P_{atm})=u^2(cal,P_{atm})$ $+u^2(dr,P_{atm})+u^2(res,P_{atm})$	
— calibration : $u(cal,P_{atm})$		$\frac{170}{2}=85$ Pa	
— drift : $u(dr,P_{atm})$		$\frac{60}{\sqrt{3}}=35$ Pa	
— reading of the barometer : $u(read,P_{atm})$		$\frac{10}{2\sqrt{3}}=2{,}89$ Pa	
	$u(P_{atm})$	91,8 Pa	$\frac{u(P_{atm})}{P_m}=\frac{91{,}8}{100281}=9{,}2.10^{-4}$

C.5.3 Estimation of the combined uncertainty

The result of the calculation of the combined uncertainty according to the equation (C.22) is:

Standard uncertainty: $u(V_{wc})$ = 1,0 % volume

$u(V_{wc,rel})$ = 5,4 % relative

Overall uncertainty: $U(V_{wc})$ = ± 2,0 % volume (k=2)

$U(V_{wc,rel})$ = ± 10,8 % relative (k=2)

If the result of the measurement is not corrected of the collection efficiency, the uncertainty shall be increased by adding the value of correction to the overall uncertainty as an error. The correction is equal to the difference between the result of the measured value and the result when collection efficiency is taken into account:

$$Corr_{\varepsilon}=\frac{\frac{(m_{wc(cs)}+m_{wc(as)})\times 100\times V_{mol(std)}}{\varepsilon\times M_{w}}}{\frac{(m_{wc(cs}+m_{wc(as)})\times 100\times V_{mol(std)}}{\varepsilon\times M_{w}}+V_{m(std)}}\times 100-\frac{\frac{(m_{wc(cs)}+m_{wc(as)})\times V_{mol(std)}}{.M_{w}}}{\frac{(m_{wc(cs)}+m_{wc(as)})\times V_{mol(std)}}{M_{w}}+V_{m(std)}}\times 100$$

$Corr_{\varepsilon}=0{,}30$ % volume

The uncertainty is equal to:

$$U(V_{wc,corr})=U(V_{wc})+Corr_{\varepsilon}$$

Standard uncertainty: $u(V_{wc,duct,corr})$ = 1,2 % volume

$u(V_{wc,duct,corr,rel})$ = 6,4 % relative

Overall uncertainty: $U(V_{wc,duct,corr})$ = ± 2,3 % volume (k=2)

$U(V_{wc,duct,,rel})$ = ± 12,5 % relative (k=2)

Annex D
(informative)

Evaluation of the method in the field

D.1 General

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

D.2 Characteristics of installations

— 1st field test: INERIS bench-loop at Verneuil en Halatte (F); the bench-loop simulates combustion or waste incineration exhaust gases. Five teams took part in the 1st field test. Double measurements were not performed simultaneously but sequentially. Five different flue gas matrices were generated. Within each matrix, two sequential measurements were performed. Two additional sequential measurements were performed in flue gas matrices where the flue gas concentrations varied. There were a total of twelve measurements performed by all the teams.

— 2nd field test: waste incinerator in Denmark. Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen measurements were performed by all the teams:

— 3rd field test: waste incinerator in Italy. Four teams took part to the field test. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements were performed by all teams.

— 4th field test: co-incinerator combined heat and power installation in Sweden. The fluidised bed boilers operate on fuel mixes of wood chips, demolition waste, peat and coal. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements were performed by each pair of two teams while a total of twelve single measurements were performed by all the teams.

— 5th field test: co-incinerator cement plant in Germany. The fuel could be coal, heavy oil and secondary fuel (e.g. paper, plastics, textiles, and tires). Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen double measurements were performed by all the teams.

— 6th field test: coal fired power plant in Germany. Four teams performed their double measurements simultaneously. The total amount of double measurements performed by all teams is 12.

An overview of the flue gas characteristics is given in Table D.1.

Table D.1 — Example of flue gas characteristics during field tests

Field test	Installation	Fuel	Flue gas characteristics						
			T °C	O_2 % vol	Nox mg/(n)m^3	SO_2 mg/(n)m^3	CO g/(n)m^3	H_2O % vol	PM mg/(n) m^3
1	Power plant[a]	Natural gas	< 150	5 to 13	10 to 1 300	10 to 2 000	20 to 400	10 to 21	<1
2	Waste incinerator	Municipal waste	90 to 110	8 to 11	180 to 250	25 to 250	5 to 15	13 to 19	1 to 5
3	Waste incinerator	Municipal waste	85v105	16 to 18	140 to 210	5 to 50	0 to 2	8 to 12	1 to 5
4	Co-incinerator	Wood, waste, peat, coal	70 to 80	4 to 6	4 to 70	0 to 10	50 to 150	8 to 12	0 to 20
5	Co-incinerator	Coal, oil, waste	140 to 170	4 to 6	440 to 1 060	60-170	260 to 740	23 to 26	5 to 10
6	Power plant	Coal	130 to 140	8,9 to 9,2	110 to 140	1 000 to 1 130	3 to 6	5,5 to 8	< 50

[a] Bench-loop: flue gas simulation.

D.3 Repeatability and reproducibility in the field

Repeatability standard deviation s_r and reproducibility standard deviation s_R are determined by performing inter-laboratory tests.

Repeatability standard deviation s_r, internal confidence interval (CI_r) and repeatability in the field r are calculated according to ISO 5725-2 and ISO 5725-6, from the results of the double measurements implemented by the same laboratory.

$$CI_r = t_{0,95;n-1} \times s_r \text{ and } r = \sqrt{2}\, .t_{0,95;n-1} \times s_r$$

where

CI_r is the internal confidence interval;

s_r is the repeatability standard deviation;

$t_{0,95;n-1}$ is the student factor for a level of confidence of 95% and a degree of freedom of n-1 (n: number of double measurements) ;

r is the repeatability in the field.

Reproducibility standard deviation s_R, external confidence interval (CI_R) and reproducibility in the field R are calculated according to ISO 5725-2, from the results of parallel measurements performed simultaneously by several laboratories.

$CI_R = t_{0,95;np-1} \times s_R$ and $R = \sqrt{2}.t_{0,95;np-1} \times s_R$

where

CI_R is the external confidence interval;

s_R is the repeatability standard deviation;

$t_{0,95;np-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of np-1 (n: number of measurements; p: number of laboratories);

R is the reproducibility in the field.

D.3.1 Repeatability

Table D.2 — Repeatability in the field

Field test	Concentration		Number of teams	Number of double measurements	Repeatability standard deviation s_r	95 % confidence interval for a single measurement CI_r		Repeatability R	
	Range (% vol.)	Average (% vol.)			% volume	% volume	% relative	% volume	% relative
1A	8,9 to 10,9	9,7	5	2	0,43	1,0	10,3	1,4	14
1B	7,3 to 9,1	8,4	5	2	0,36	0,8	9,6	1,1	13
1C	10,7 to 13,1	11,8	5	2	0,55	1,2	10,2	1,8	15
1D	13,1 to 16,0	14,4	5	2	0,69	1,6	10,5	2,2	15
1E	22,0 to 25,1	22,7	5	2	1,17	2,6	11,5	3,7	16
1F	15,4 to 20,3	17,7	5	1	0,88	2,4	13,6	3,5	20
1G	15,0 to 20,3	17,2	5	1	0,86	2,4	13,9	3,4	20
2	13,9 to 18,2	17	4	16	0,81	1,7	10,0	2,5	14
3	8,2 to 11,7	10,4	4	12	0,57	1,3	12,5	1,8	17
4	8,5 to 11,6	9,6	4	12	0,56	1,2	12,5	1,7	18
5	21,2 to 25,7	22,5	3	12	0,56	1,3	5,8	1,8	8
6	5,5 to 7,8	6,0	3	12	0,16	0,35	5,8	0,5	8

The following functions were determined:

s_r(C)= 0,0218 C + 0,247;

$s_{r\ limit}$(C)= 0,033 C + 0,37;

CI_r(C)= 0,05 C + 0,5;

r(C)= 0,071 C + 0,7.

D.3.2 Reproducibility

Table D.3 — Reproducibility in the field

Field test	**Concentration**		**Number of teams**	**Number of double measurements**	**Reproducibility standard deviation** s_R	**95 % confidence interval for a single measurement** CI_R		**Reproducibility** R	
	Range (% vol.)	Average (% vol.)			% volume	% volume	% relative	% volume	% relative
1A	8,9 to 10,9	9,7	5	2	0,83	1,9	19,6	2,7	27
1B	7,3 to 9,1	8,4	5	2	0,55	1,2	14,3	1,8	21
1C	10,7 to - 13,1	11,8	5	2	0,77	1,7	14,4	2,5	21
1D	13,1 to 16,0	14,4	5	2	1,00	2,4	16, 7	3,4	23
1E	22,0 to 25,1	22,7	5	2	1,70	3,8	16,8	5,4	24
1F	15,4 to 20,3	17,7	5	1	2,00	5,4	30,5	7,7	44
1G	15,0 to 20,3	17,2	5	1	2,10	5,8	33,8	8,3	48
2	13,9 to 18,2	17	4	16	0,99	2,1	12,4	3,0	18
3	8,2 to 11,7	10,4	4	12	0,67	1,5	14,4	2,1	20
4	8,5 to 11,6	10,1	4	12	0,72	1,6	15,8	2,3	22
5	21,2 to 25,7	22,5	3	12	0,96	2,1	9,3	3,0	13
6	5,5 to 7,8	6,0	3	12	0,48	1,1	18,3	1,5	25

The following functions were determined:

S_R(C)= 0,03 C + 0,367 in %;

CI_R(C)= 0,066 C + 0,9 in %;

R(C)= 0,093 C + 1,27 in %.

Annex ZA
(informative)

Relationship between this European Standard and the Essential Requirements of EU Directive

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of the EU Council Directive 2000/76/EC on the Incineration of Hazardous Waste and of the EU Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.

WARNING — Other requirements and EU Directives may be applicable falling within the scope of use of this European Standard.

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